SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Scenario-Based

Course Code:	ITA0532	**Course Title:**	Computer Vision
CO Assessed:	CO2 – Imitation (BL3)	**Assessment:**	Assessment Tool 1 – **Scenario-Based**
Weightage:	10% of CIA	**Total Marks:**	100
CO2 Statement:	*Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*		
Student Name:		**Reg. No.:**	
Section / Batch:		**Date:**	

Code of Conduct

I ______________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ______________________________

ASSESSMENT TOOL 2: Scenario-Based

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Scenario	20%					
Identification of Key Issues	20%					
Application of Concepts	25%					
Decision Making	20%					
Clarity of Response	15%					

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Question 6: Object Boundary Detection

Scenario: *In a medical image, boundaries between tissues need to be clearly identified.*

(a) Interpretation of the Scenario — Importance of Boundary Detection

In medical imaging (CT, MRI, ultrasound, X-ray), a tissue boundary is the line where one anatomical structure ends and another begins — for example, where a tumour meets healthy tissue, or where an organ meets surrounding fat or muscle. Detecting these boundaries accurately is not a cosmetic step; it is the foundation on which every downstream clinical decision is built.

This scenario matters because:

- **Diagnosis depends on shape and extent:** A radiologist judges malignancy, growth pattern, and severity largely from the shape, smoothness, and size of a boundary. A blurred or wrongly-detected edge can hide or exaggerate a problem.
- **Surgical and treatment planning:** Surgeons and radiotherapy planners use the exact boundary to decide how much tissue to remove or how to aim a radiation beam, so that healthy tissue is spared.
- **Quantitative measurement:** Tumour volume, organ size, and growth over time (across scans) are all calculated by measuring the region enclosed by the detected boundary.
- **Automated diagnosis pipelines:** Boundary detection is usually the first step before segmentation and classification in AI-assisted diagnosis systems — errors here propagate through the whole pipeline.

In short, boundary detection converts a raw grayscale scan into clinically meaningful geometric information, so its accuracy is directly tied to patient safety and treatment quality.

(b) Key Challenges in Detecting Boundaries

Medical images are far harder to work with than typical photographs. The main challenges are:

Challenge	Why it happens	Effect on Boundary Detection
Low contrast between tissues	Soft tissues often have very similar X-ray/MRI absorption or echo properties	Edges are weak — true boundary can be missed
Noise (speckle / quantum noise)	Ultrasound speckle, X-ray photon noise, MRI thermal noise	False edges appear; real edge gets buried in noise
Blurred / fuzzy edges	Partial volume effect — one pixel covers more than one tissue type	Edge location becomes uncertain, not a sharp line
Irregular, non-uniform shapes	Tumours and organs rarely have simple geometric shapes	Simple/rigid detection methods fail to trace the true outline
Artifacts	Patient motion, metal implants, scanner imperfections	Creates fake edges unrelated to real anatomy
Need for sub-pixel accuracy	Even a few pixels of error changes measured volume/area	Ordinary thresholding is not precise enough

These challenges mean that a naive method (like simple thresholding) is unreliable — the chosen technique must be robust to noise while still being sensitive enough to catch faint, real boundaries.

(c) Applying a Suitable Edge Detection Method — Canny Edge Detector

The Canny Edge Detector is selected as the suitable method for this scenario. It is a multi-stage algorithm, and each stage directly targets one of the challenges identified in part (b):

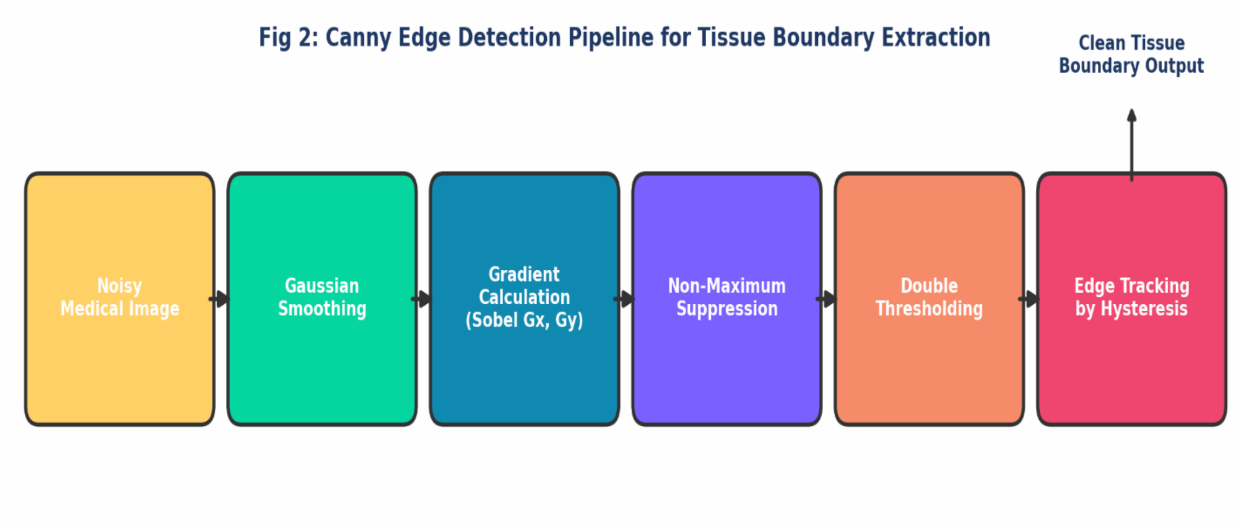

Fig 2: Canny edge detection pipeline for tissue boundary extraction

Stage	What it does	Challenge it solves
1. Gaussian Smoothing	Blurs the image slightly with a Gaussian kernel to suppress noise	Reduces speckle / quantum noise
2. Gradient Calculation (Sobel)	Computes intensity gradient magnitude and direction at every pixel	Detects weak, low-contrast edges via rate of change
3. Non-Maximum Suppression	Keeps only the local maximum gradient along the edge direction	Turns thick, blurred edges into a thin, precise line
4. Double Thresholding	Classifies pixels as strong / weak / non-edge using two thresholds	Separates real anatomical edges from artifact noise
5. Edge Tracking by Hysteresis	Keeps weak edges only if connected to a strong edge	Recovers faint but genuine boundary segments

Applied to a simulated low-contrast, noisy tissue image (grey-level difference of only ~28 between tissue and background, with added Gaussian noise), the pipeline performs as follows:

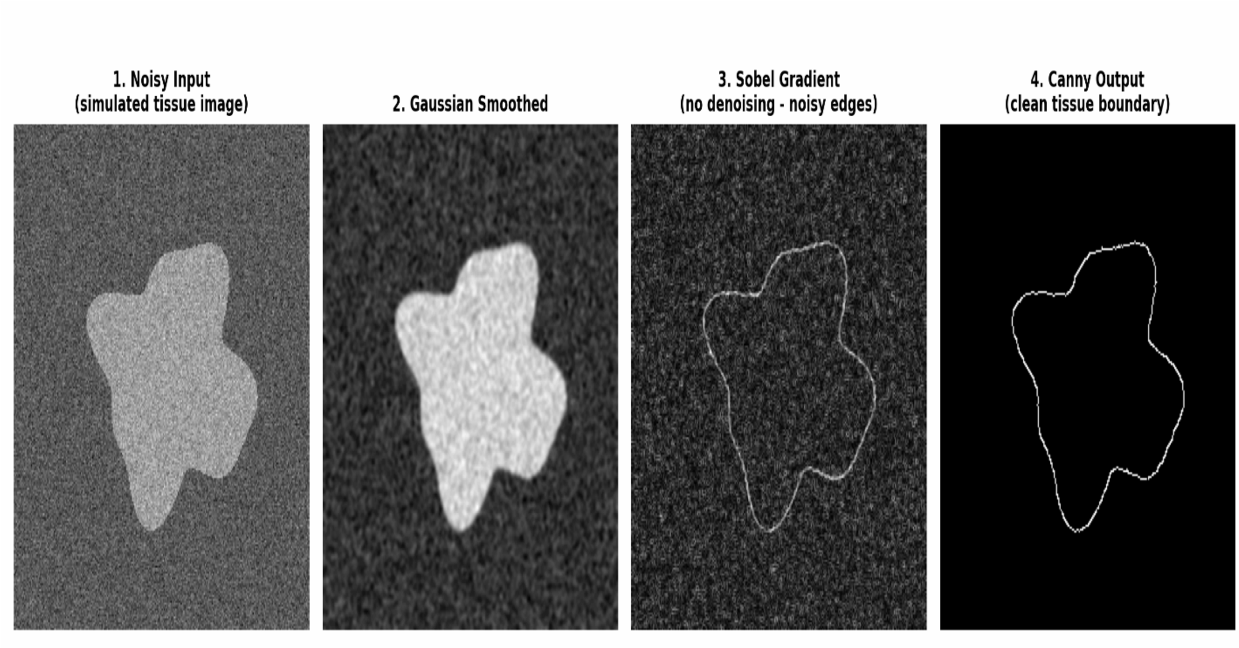

Fig 1: Real OpenCV output — noisy input, smoothing, plain Sobel gradient, and final Canny boundary

As the output shows, a plain gradient method (Sobel, panel 3) still carries visible noise texture along the edge, while the full Canny pipeline (panel 4) produces a single clean, continuous, one-pixel-wide boundary that traces the exact irregular shape of the tissue region.

(d) Justification of the Selection — Decision Making

The choice of Canny over other common edge detectors is justified by direct comparison:

Method	Noise Handling	Edge Thickness	Sensitivity to Faint Edges	Verdict for this Scenario
Sobel / Prewitt	Poor — no built-in smoothing	Thick, imprecise	Moderate	Rejected — noisy medical images give ragged edges
Laplacian of Gaussian (LoG)	Good (Gaussian built-in)	Can double-detect edges	High, but noisy in flat regions	Possible, but less precise localisation than Canny

Simple Thresholding	None	N/A (region-based)	Fails on low contrast	Rejected — cannot separate similar-intensity tissues
Canny Edge Detector	Strong (Gaussian + hysteresis)	Thin, single-pixel	High, with false-edge rejection	Selected — best balance for this scenario

The deciding factors are:

- **Robust noise suppression:** The Gaussian smoothing stage directly counters the speckle/quantum noise identified as a key challenge, without needing a separate pre-processing step.
- **Precise, thin boundaries:** Non-maximum suppression gives a one-pixel-wide edge, which is essential for accurate area/volume measurement in diagnosis.
- **Hysteresis recovers faint real edges:** Because tissue boundaries are often low-contrast, the dual-threshold + connectivity check keeps genuine weak edges that a single-threshold method would discard, while still rejecting isolated noise pixels.
- **Widely validated in medical imaging:** Canny is a standard first step in many clinical image-processing pipelines because it consistently balances sensitivity and false-edge rejection better than single-stage gradient methods.

Hence, for a scenario demanding accurate, noise-resistant, sub-pixel tissue boundaries, Canny is the logically superior choice over Sobel, Prewitt, LoG, or simple thresholding.

(e) Clear and Systematic Presentation

The complete reasoning for this scenario can be summarised in one flow: a diagnostic need for accurate tissue boundaries leads to specific challenges (noise, low contrast, blur, irregular shape), which are addressed by choosing a method whose stages are individually matched to those challenges, and validated by comparing it against simpler alternatives.

Step	Summary
Scenario	Medical image with two tissue regions needing a clear boundary
Key Challenges	Low contrast, noise, blurred edges, irregular shape, artifacts
Method Applied	Canny Edge Detector (5-stage pipeline)
Justification	Best noise-robustness + thin, precise, sub-pixel boundaries vs. Sobel / LoG / thresholding
Result	Clean, continuous, single-pixel tissue boundary suitable for diagnosis and measurement

This structured approach — interpret, identify, apply, justify, present — mirrors exactly how a clinical image-processing engineer would reason through a real diagnostic imaging task.